PERIODONTAL LIGAMENT

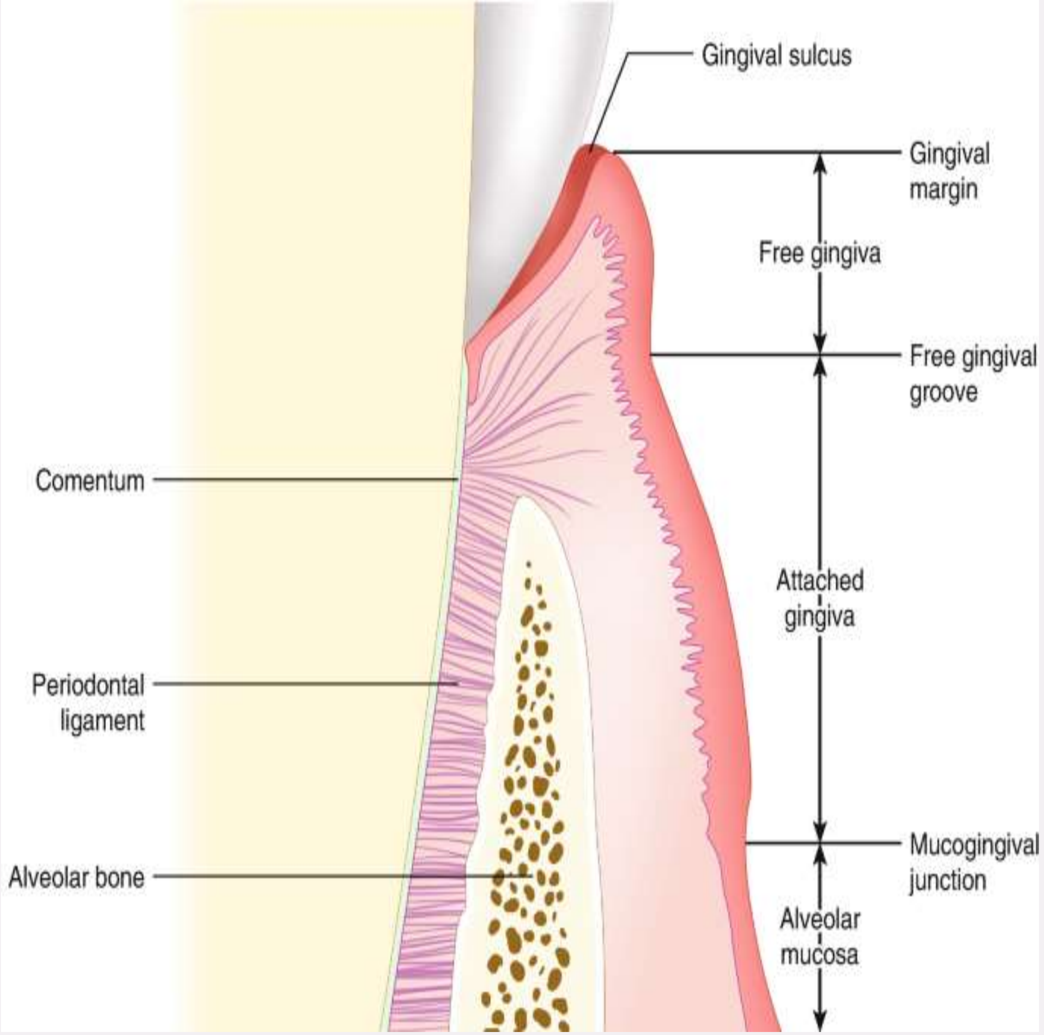
DR. UME HANI

DEPT OF ORAL BIOLOGY

- It is derived from the dental follicle.
- Dense fibrous connective tissue.
- Occupies the PDL space between the root

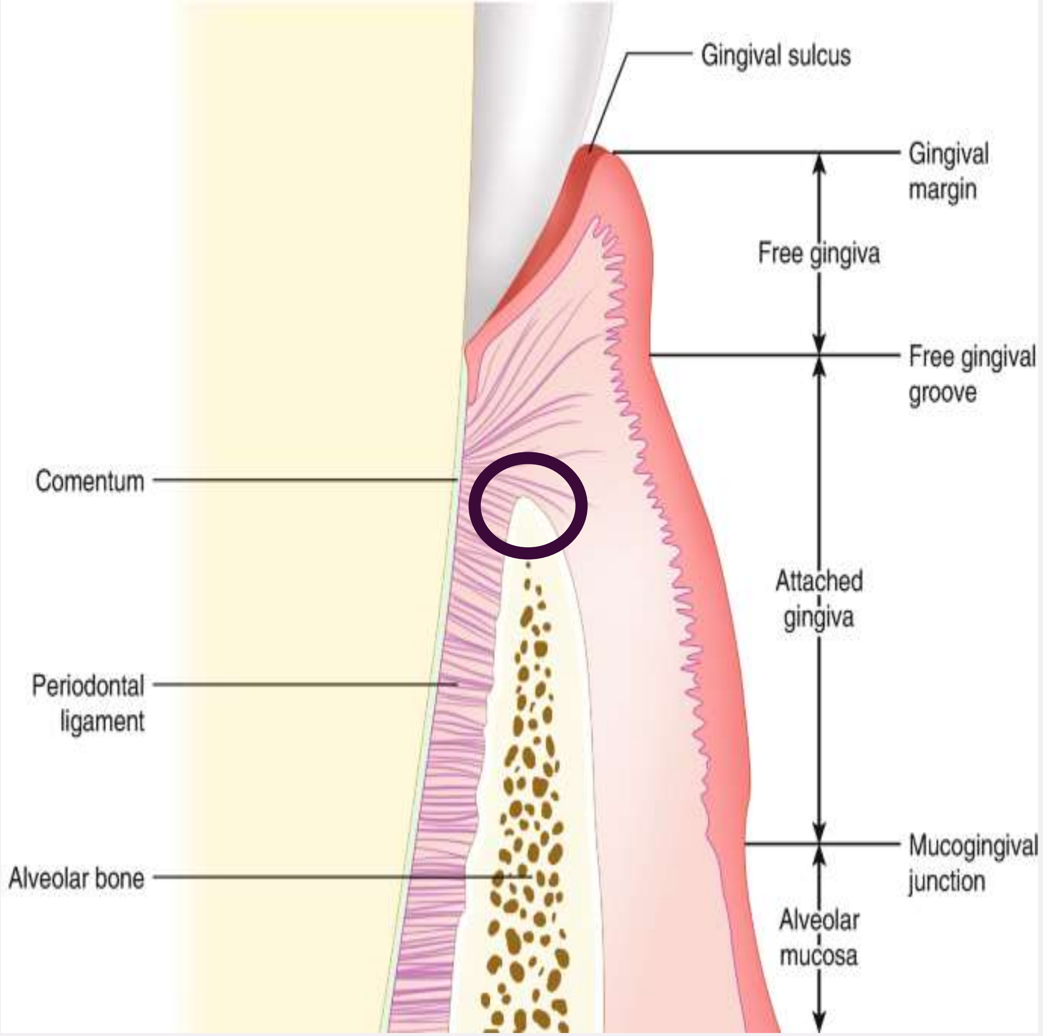
of the tooth and alveolus.

- The average width of PDL space is 0.25 mm, hourglass in shape, narrowest in the mid-root.



Above the alveolar crest, PDL is continuous with the connective tissues of the gingiva

Clinical relation: This facilitates the progression of gingivitis from periodontitis



COMPOSITION OF PDL

Cellular Components

- Synthetic Cells
- Resorptive Cells
- Epithelial Cell Rests
- Defense Cells

Extracellular Components

Stroma of Fibers

Ground Substance

Water

Stroma of periodontal fibers

- 90% collagen
- 3% Oxytalan
- Reticulin
- Elaunin

FIBRES OF PDL

- **COLLAGEN**
  - **Type I (70%)**
    - **Most abundant,provides tensile strength**
  - **Type III (20%)**
    - **Covalently linked to type I collagen**
    - **Found in periphery of sharpey's fibre attached into alveolar bone.**
    - **And around nerves and blood vessels**
    - **Reticular fibres, helps in remodelling**

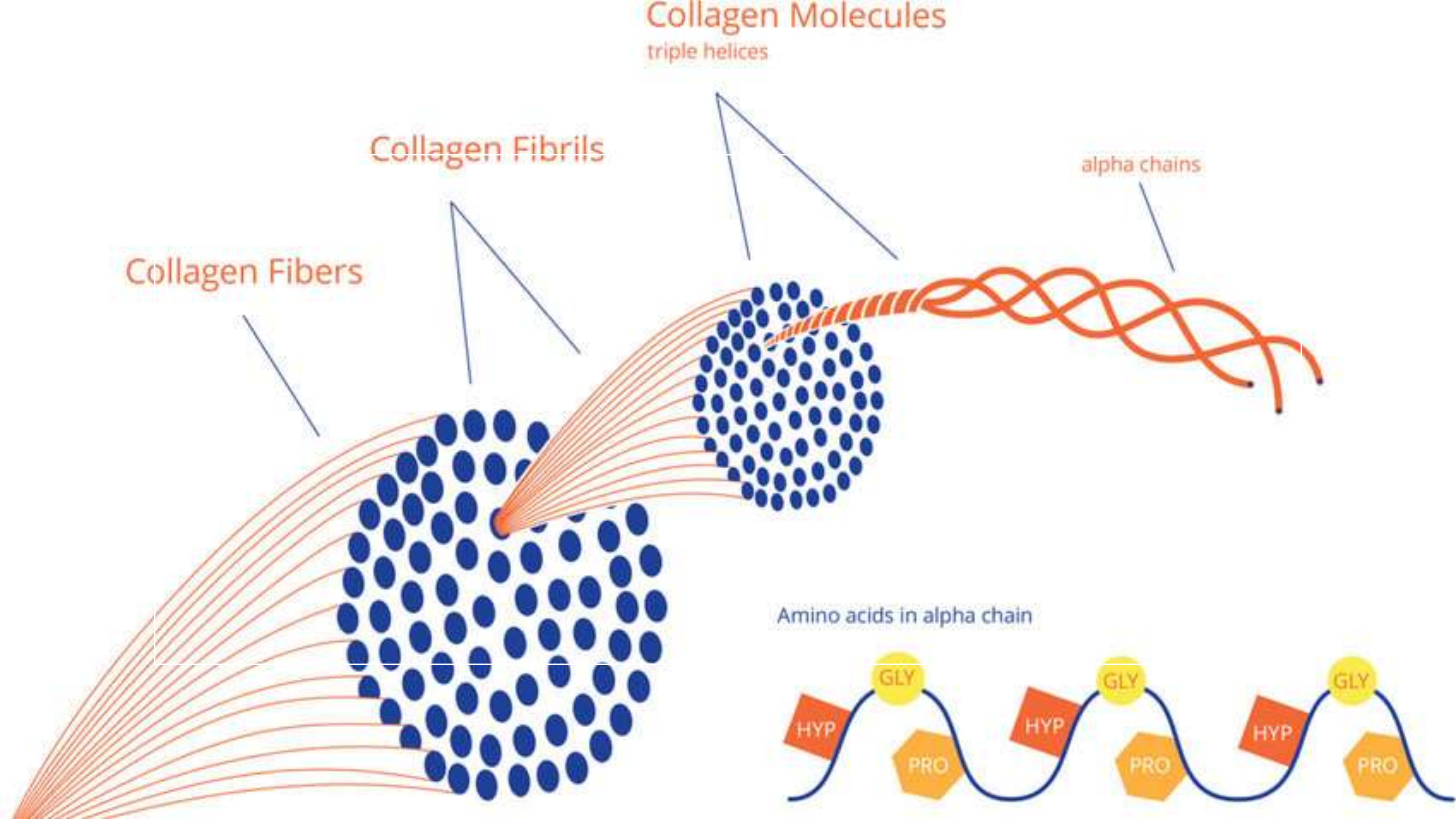
Traces of collagen

- Type V (coats cell surfaces) and VI and XIV, present in PDL
- Type IV and VII basement membrane collagens associated with epithelial cell rests & blood vessels
- Type XII collagen, a non-fibrous collagen known as fibril associated collagen regulate PDL cells for architecture of PDL

Much of the collagen is gathered to form bundles approximately "5 μm" in diameter. These bundles are termed the 'principal fibers'

The fibroblasts are responsible for the synthesis and degradation of collagen

Within each collagen bundle, subunits of a structure are called collagen fibrils


Collagen Molecules
triple helices
Collagen Fibrils
alpha chains
Collagen Fibers
Amino acids in alpha chain
GLY
HYP
PRO
GLY
HYP
PRO
GLY
HYP
PRO

Three polypeptide α-chains wind around to form a triple helix: the tropocollagen molecule.

Staggering and cross-linking pattern to form collagen fibrils.

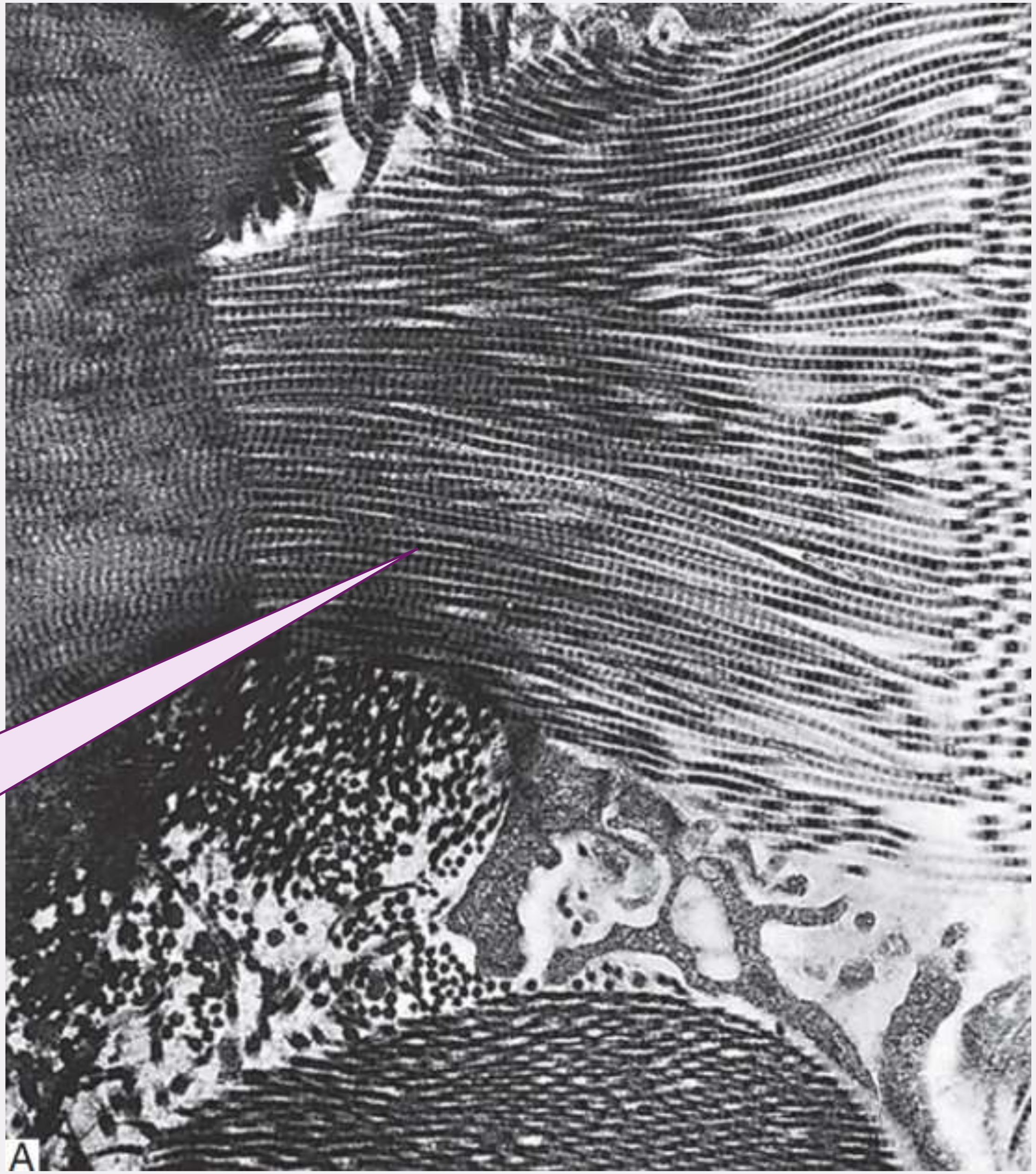
Collagen fibrils are bundled together to form a larger collagen fiber.

Collagen fibrils are bundled together to form a larger collagen fiber.

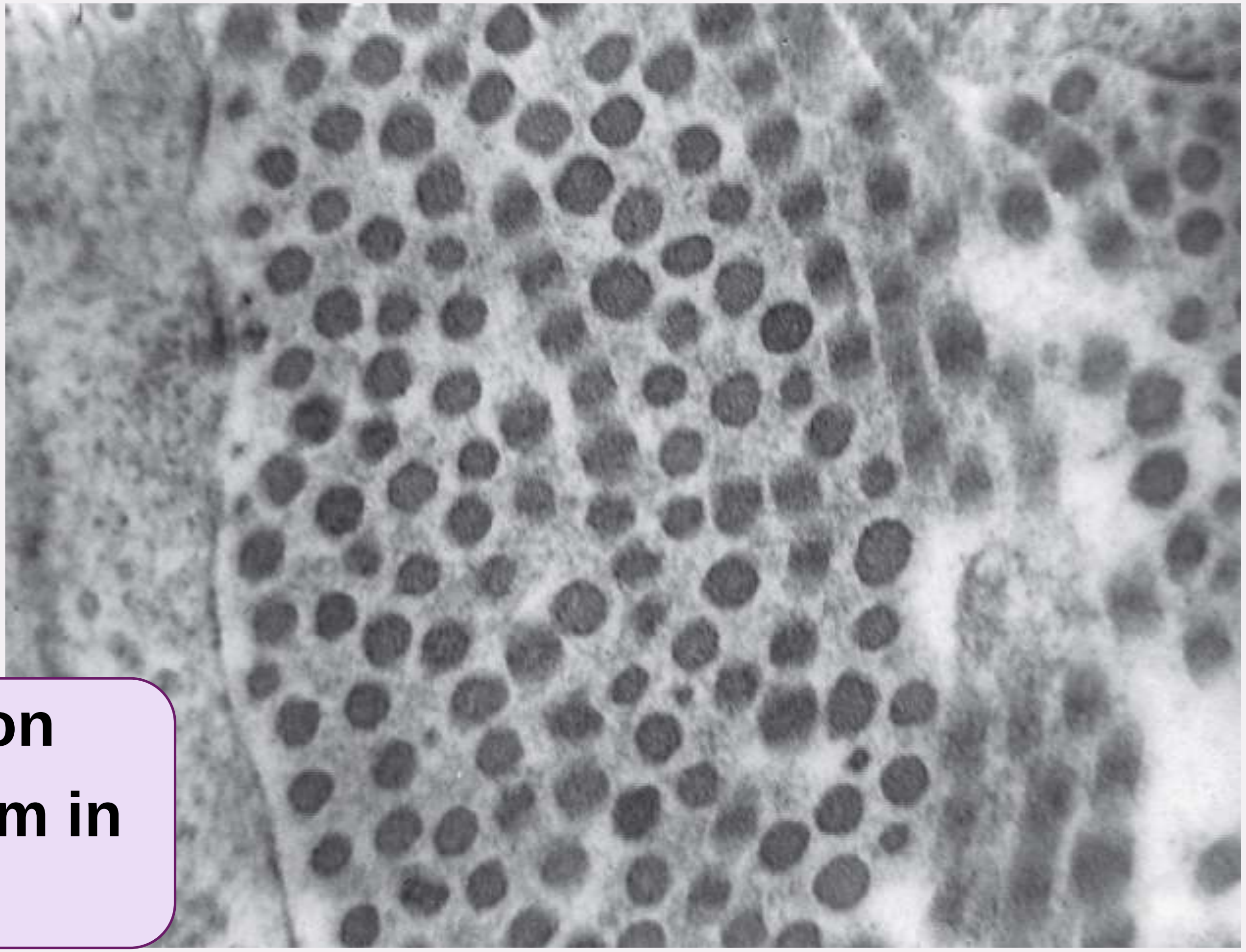
COLLAGEN FIBRILS

- The collagen fibrils are subunits within a collagen fibre bundle & formed by the packing together of individual tropocollagen molecules.
- The collagen fibrils of the periodontal ligament are small and of uniform diameter.

These fibrils in longitudinal section display the banding characteristic of collagen.



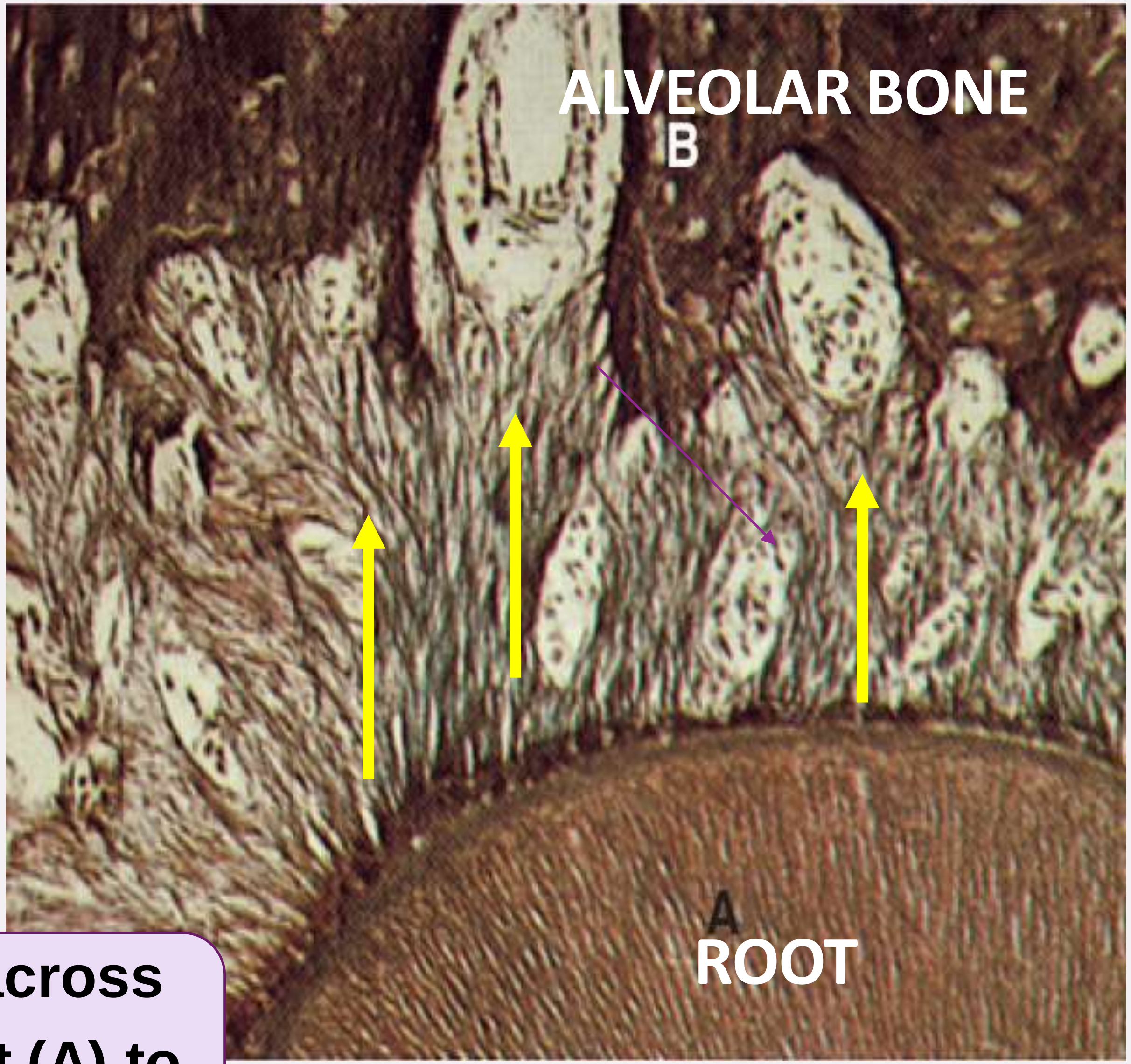
COLLAGEN FIBRILS



The fibrils in transverse section appear to be small and of uniform in diameter

PRINCIPAL FIBERS

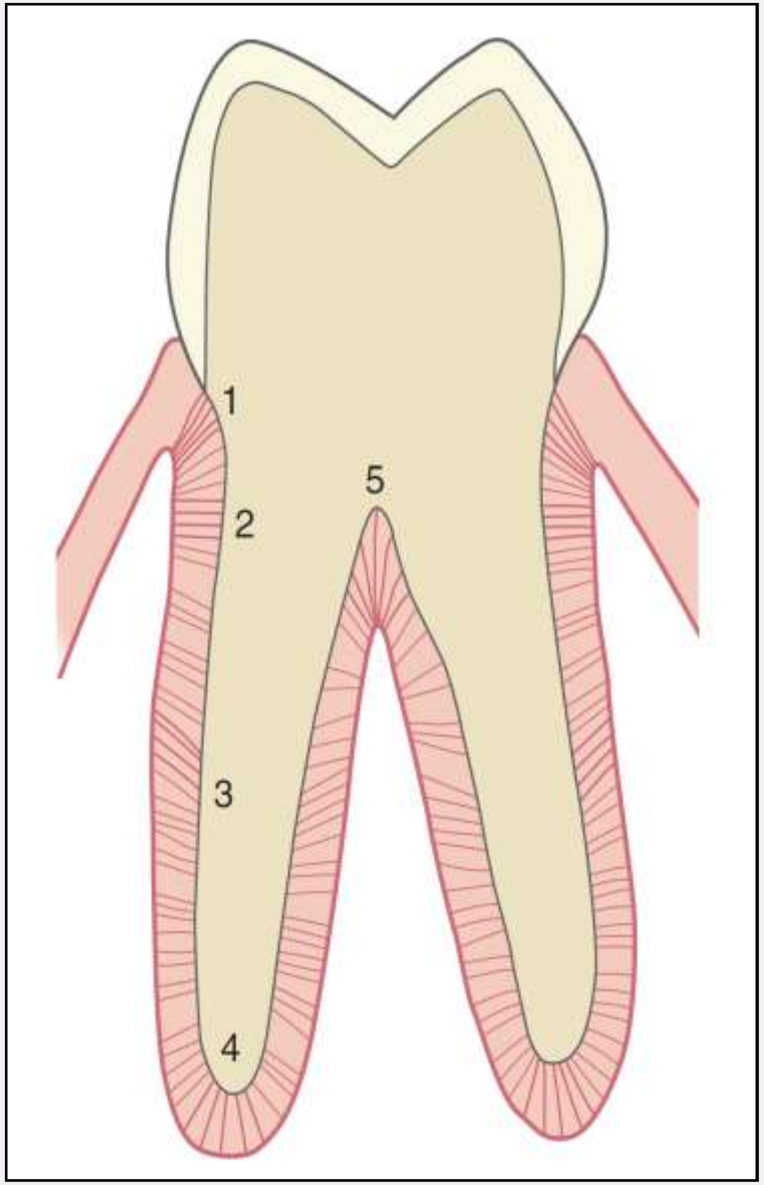
The collagen fibrils subunits of collagens fibers gathered to form bundles known as principal fibers having approx. 5 μm diameter.



Principal collagen fibres passing across the periodontal space from the root (A) to the alveolar bone (B).

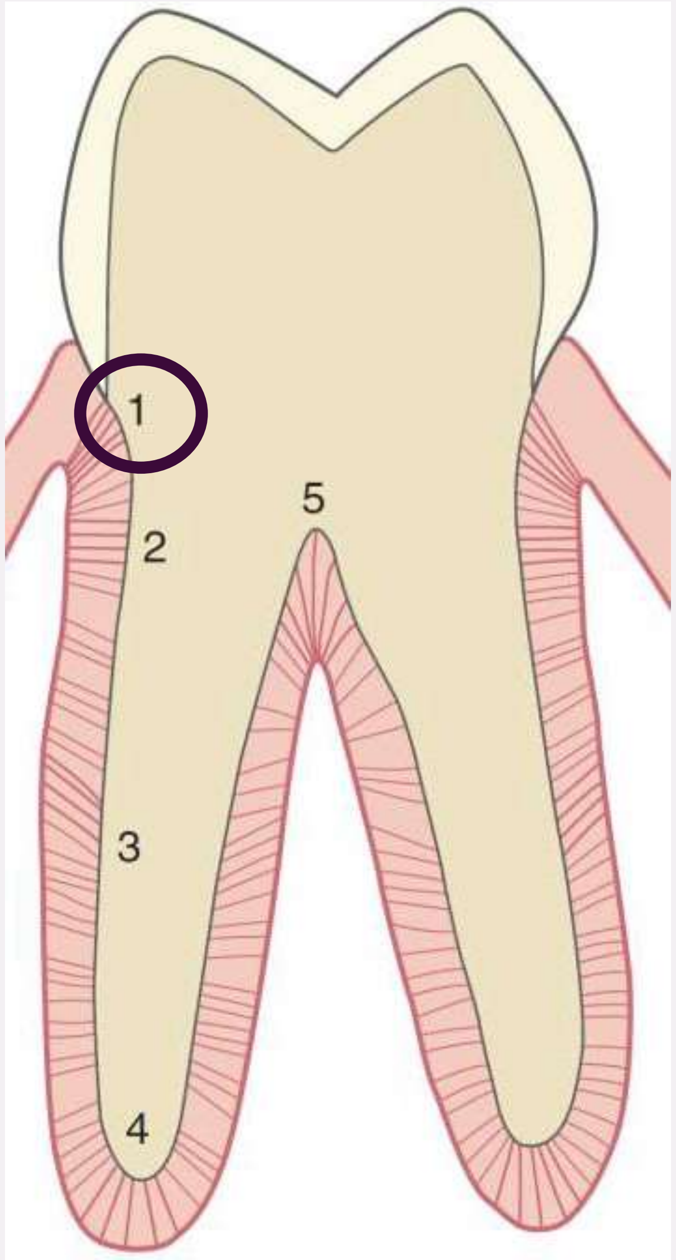
THE ORIENTATION OF THE PRINCIPAL FIBERS OF PDL SEEN IN LONGITUDINAL SECTION OF MULTIROOTED TOOTH

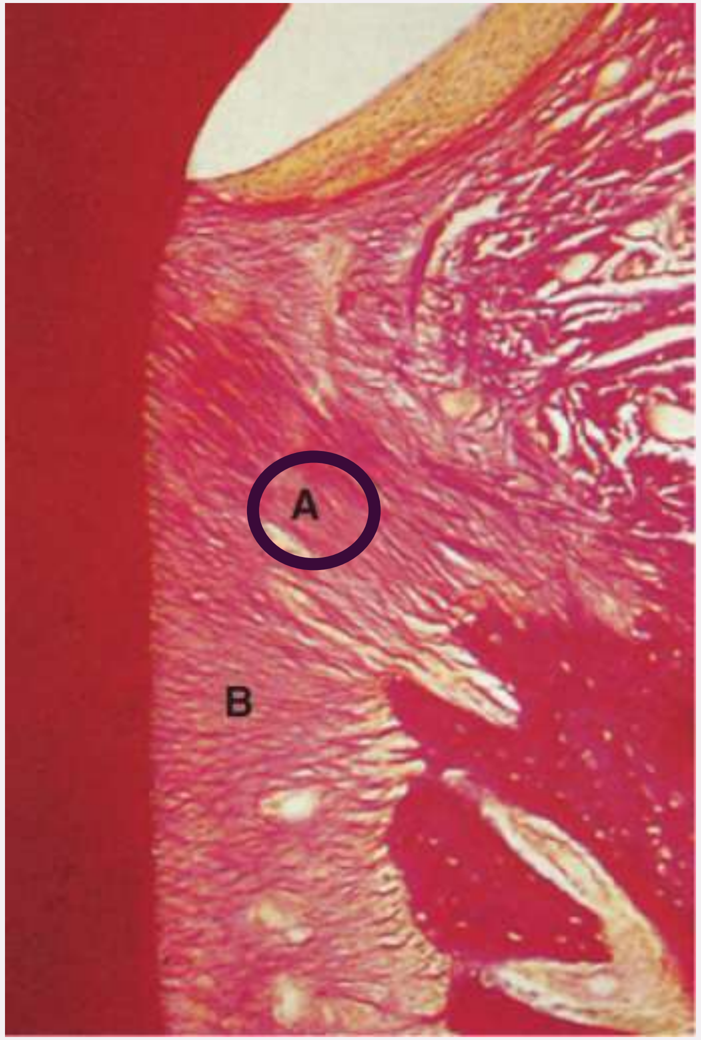
1. **Dentoalveolar Crest Fibers;**
2. **Horizontal Fibres;**
3. **Oblique Fibres;**
4. **Apical Fibres;**
5. **Interradicular Fibres.**



Dentoalveolar crest fibers:

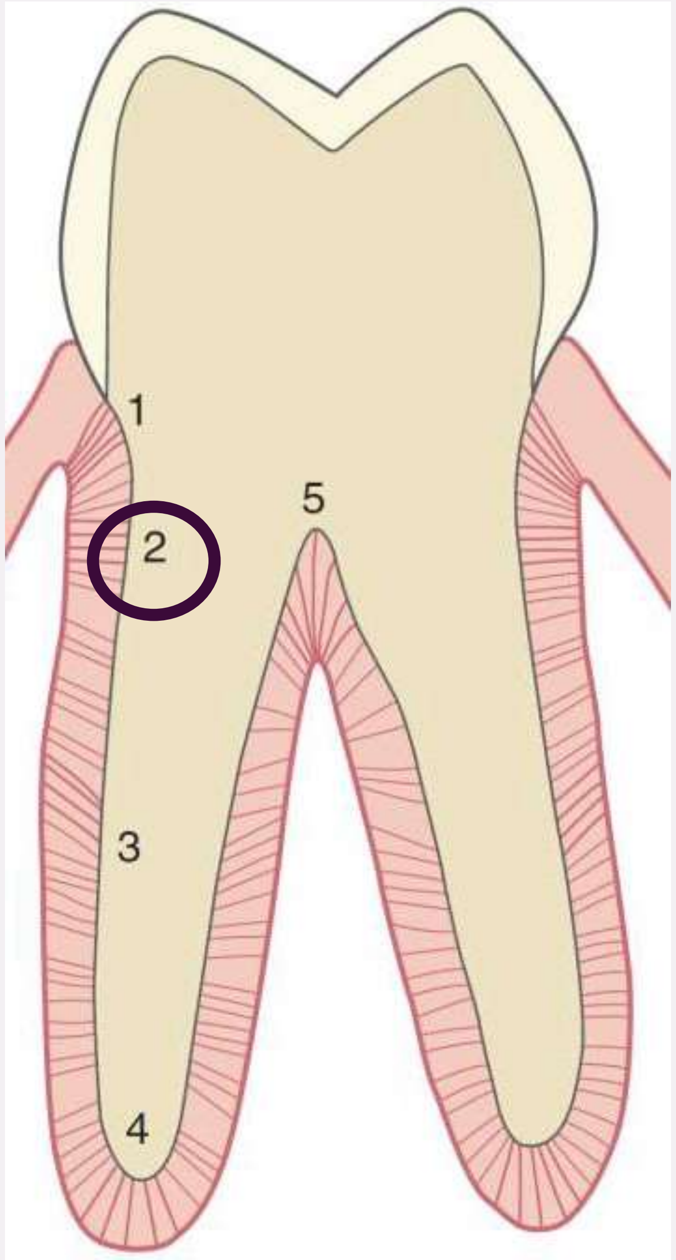
Run from the cementum just beneath the alveolar crest to the alveolar bone.

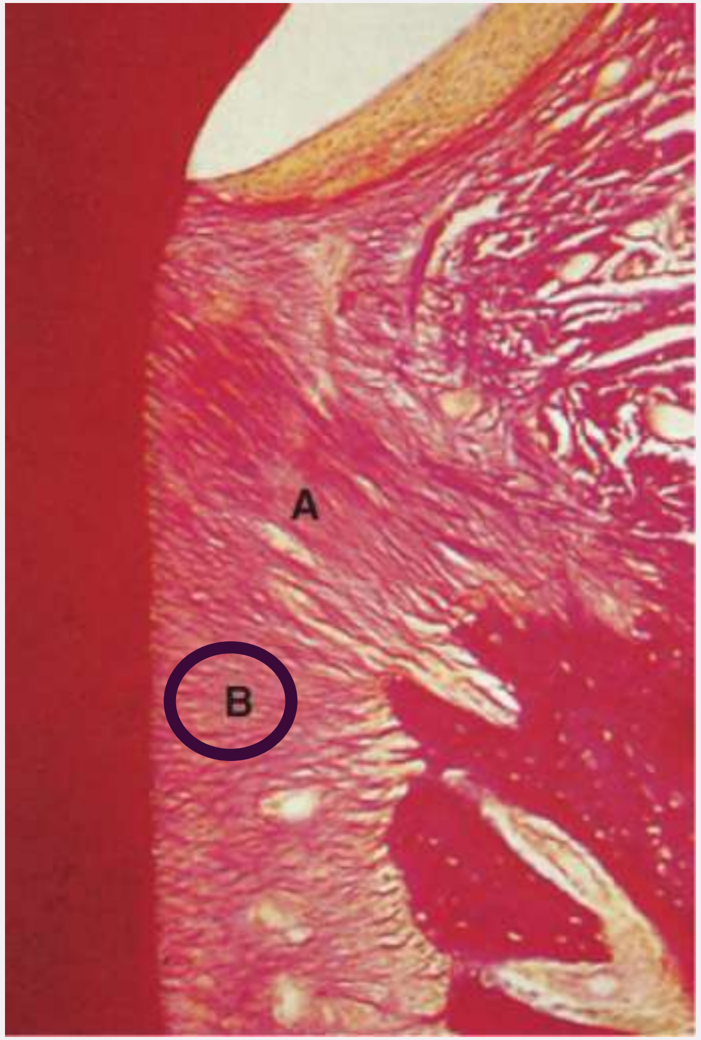




Horizontal fibers:

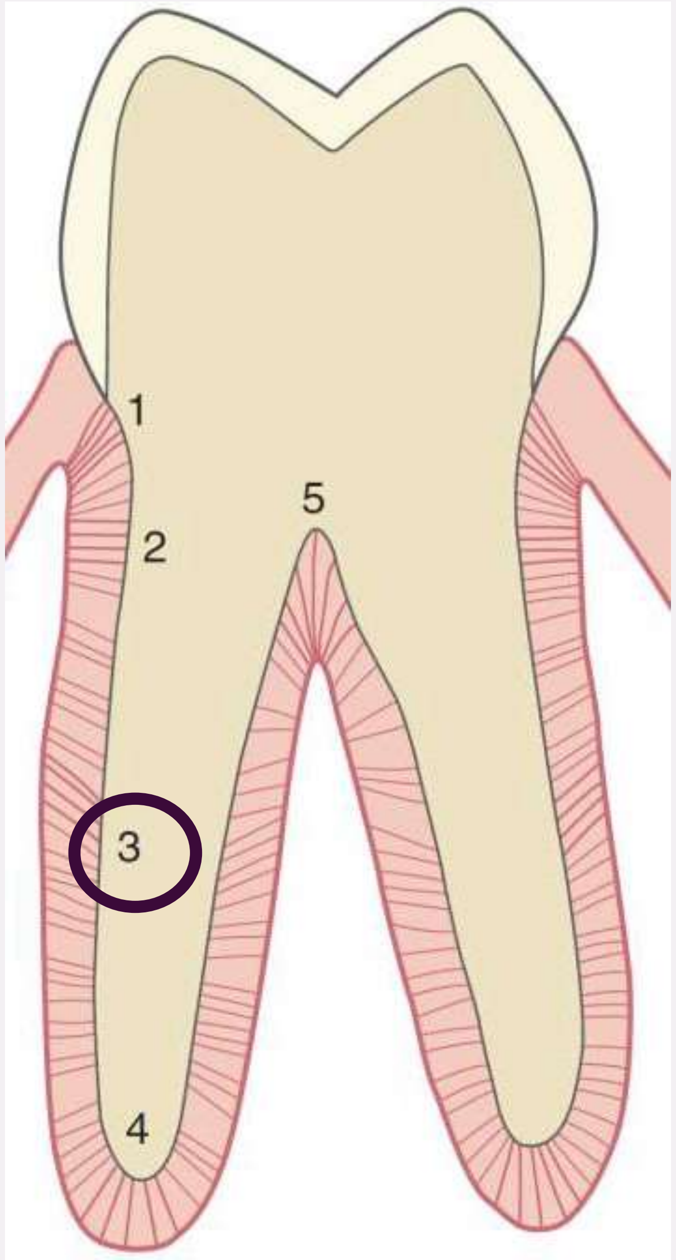
Extend horizontally from the cementum to the alveolar bone.





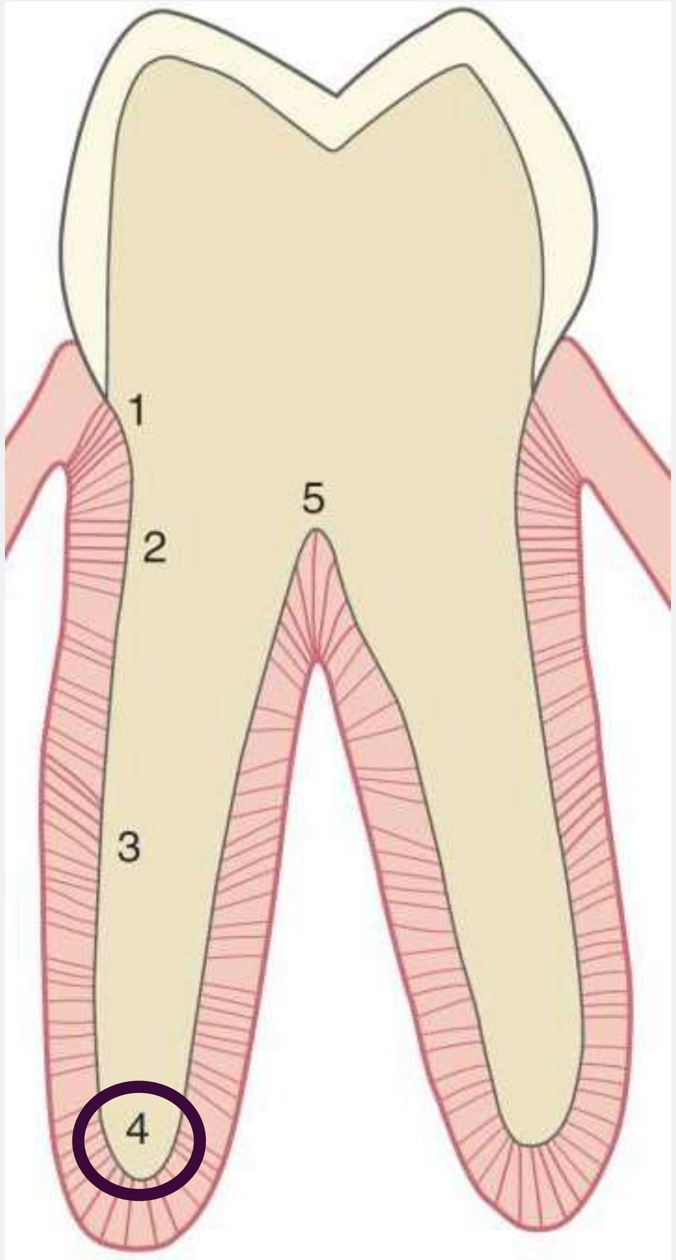
Oblique fibers:

Run obliquely (downward direction) from cementum to the alveolar bone supporting the tooth against masticatory forces.



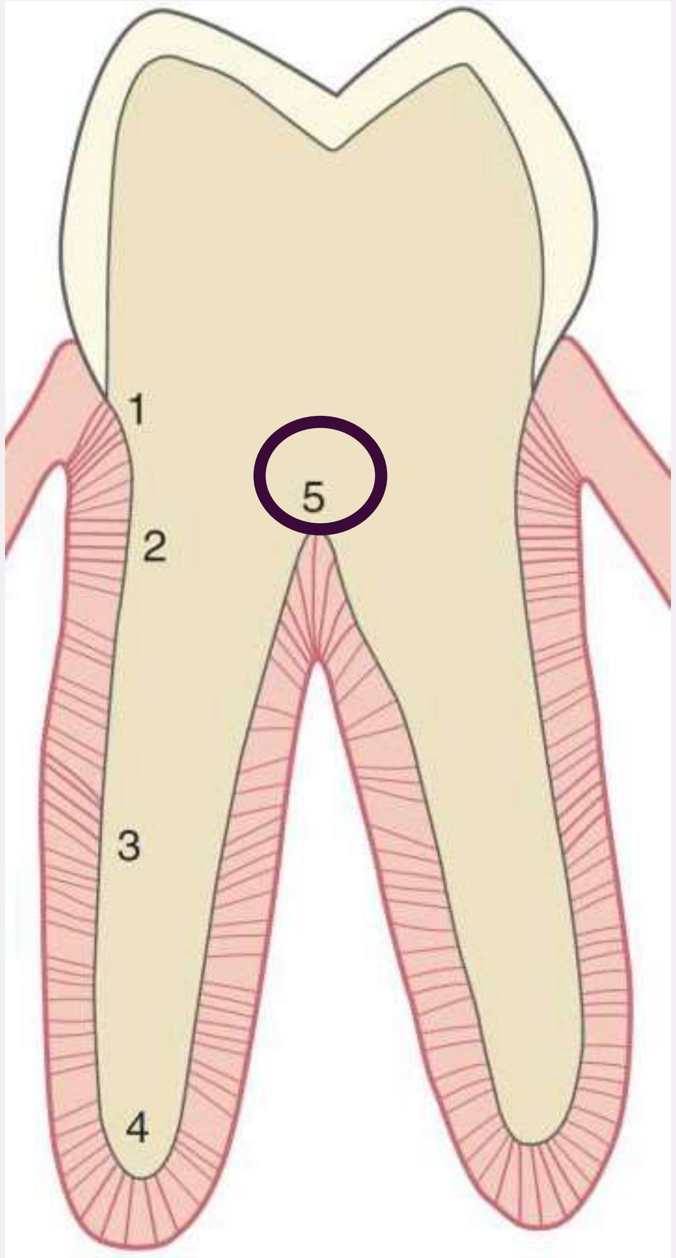
Apical fibers:

Extend from the apex, down to the alveolar bone.



Interradicular fibers:

Located only in multi-rooted teeth, fibers extend from cementum between the roots of the teeth to the interradicular septum of the alveolar bone.

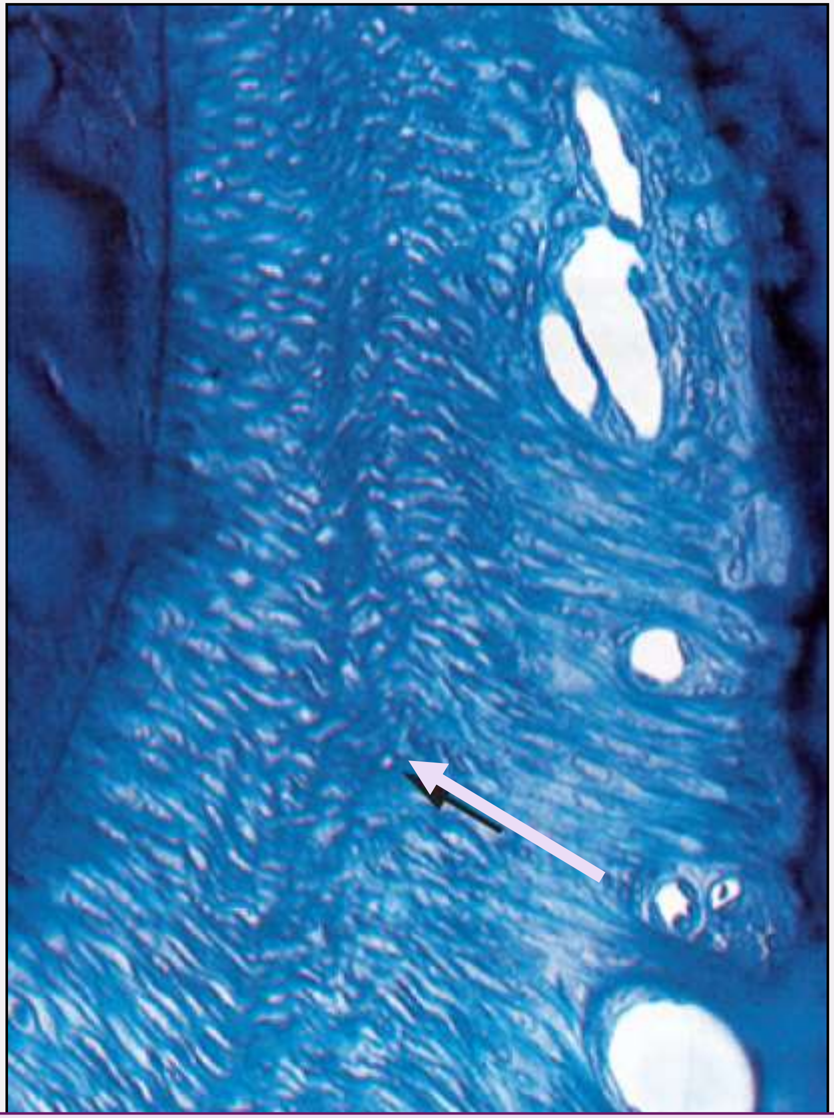


ZONE OF SHEAR;
A SITE OF REMODELLING DURING ERUPTION

A specific region in the PDL where intense remodeling occurs- midway between the cementum and alveolar bone, especially during tooth eruption and orthodontic movement.

-adaptation of fibres

-high collagen synthesis



Longitudinal section through the periodontal ligament

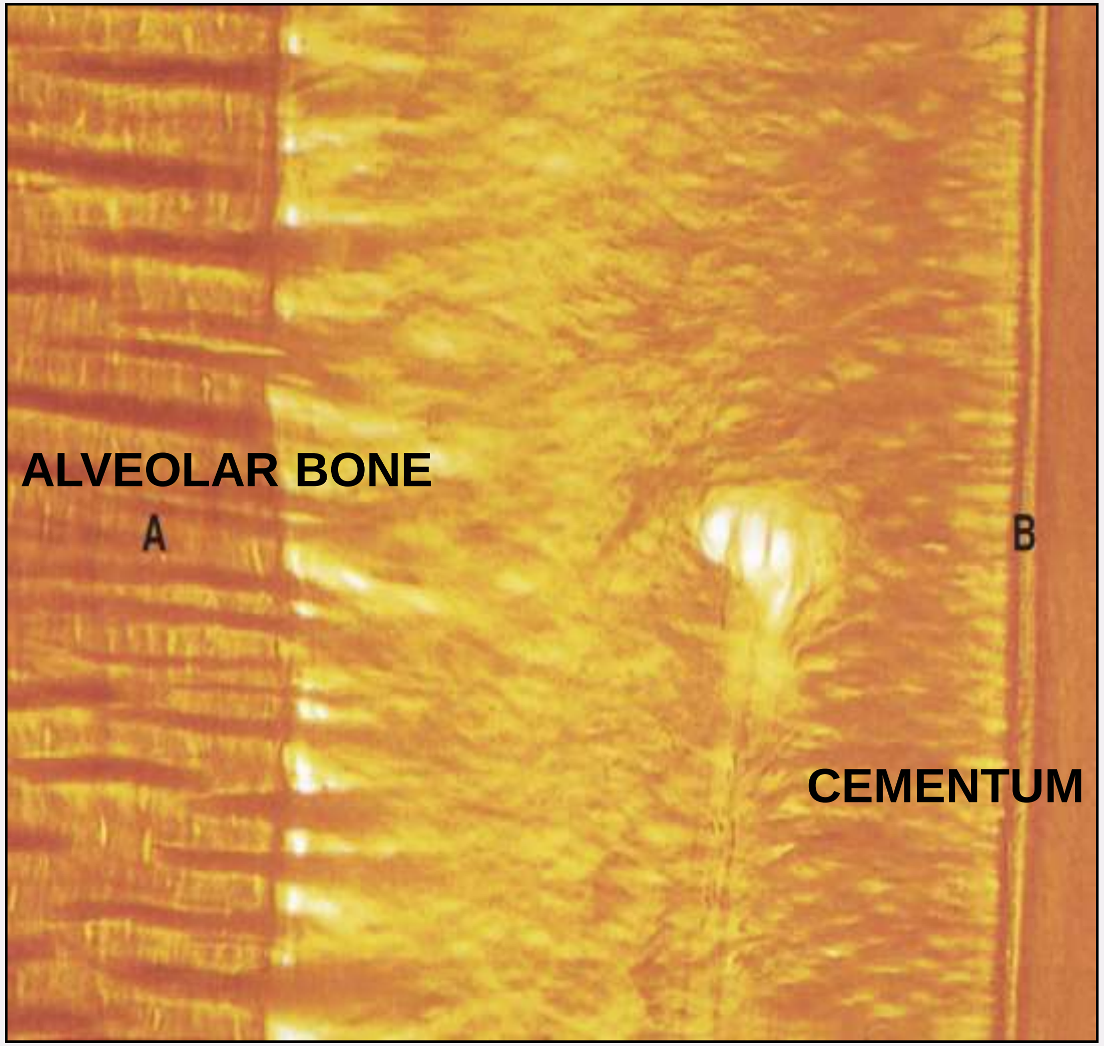
CRIMPING

The principal fibres of the pdl do not run a straight course as they pass from the region of the alveolar bone to the tooth. A specific type of waviness seen in PDL is crimping.

(Transverse section) crimping of collagen fibres is seen to be banded between crossed polarized light, reflecting an underlying periodicity

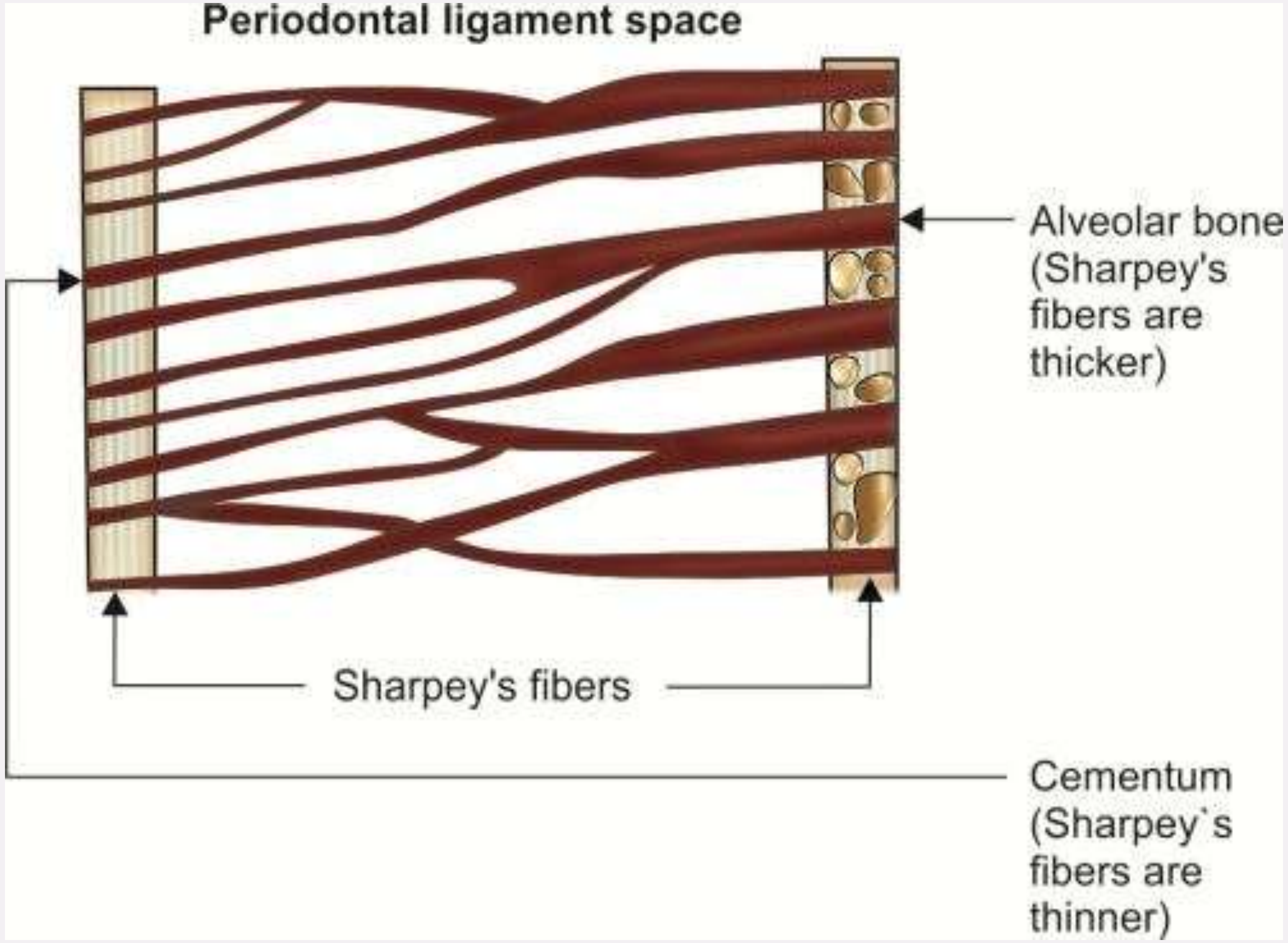
SHARPEY'S FIRBERS

The principal fibres of the periodontal ligament that are 'embedded' into cementum and the bone lining the tooth socket are termed Sharpey's fibres.



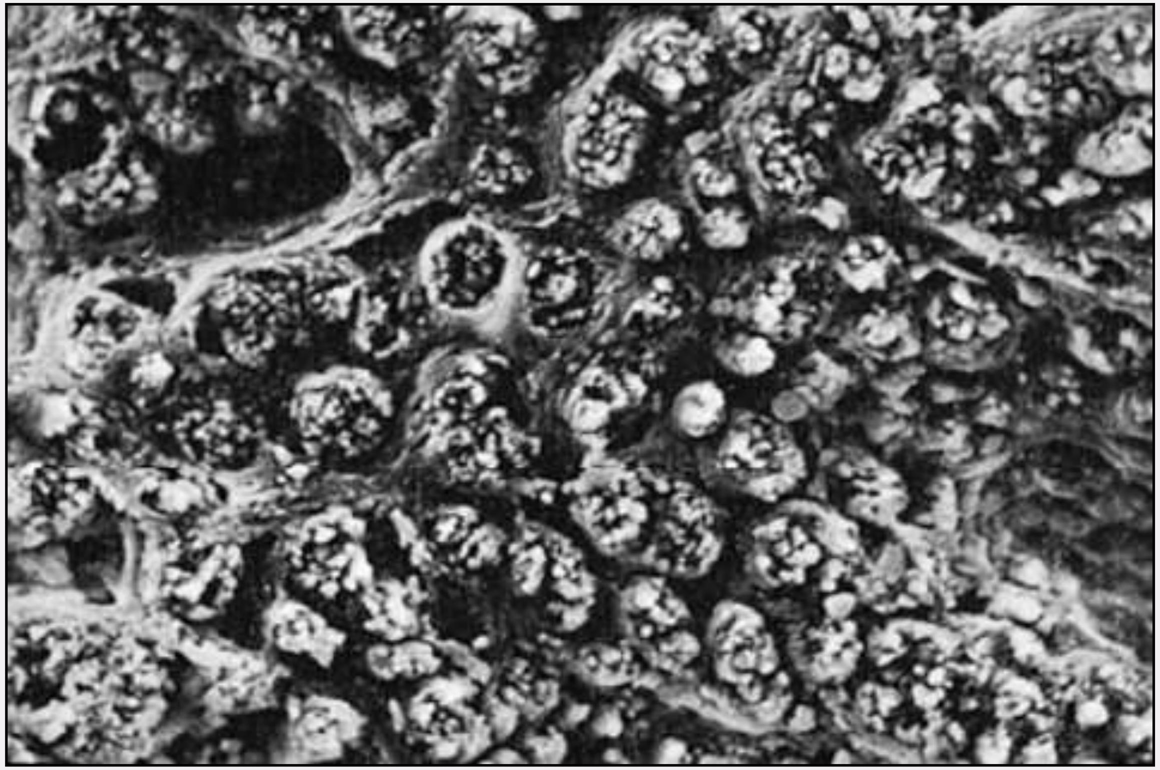
The horizontal lines in the bone and cementum represent the Sharpey's fibres.

These Sharpey's fibres are more numerous, but smaller, at their attachments into cementum than at the alveolar bone.



SHARPEY'S FIBERS

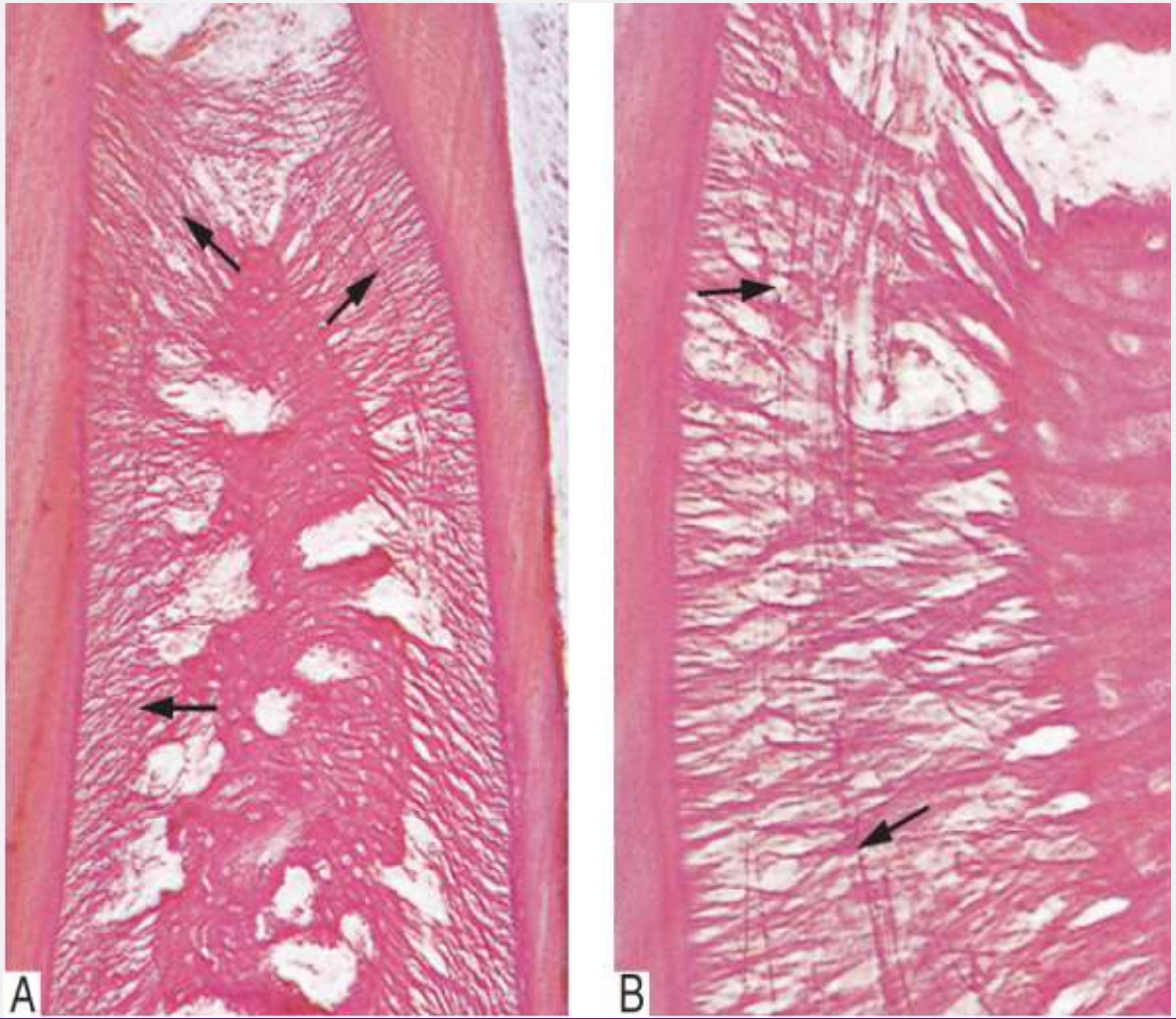
Mineralised in alveolar bone so appear as projecting stubs covered with mineral clusters.



SEM of Sharpey's fibres in the alveolar bone showing 'stublike' appearance

OXYTALAN FIBERS

- Immature elastin fibres (pre-elastin) and contain microfibrils.
- Attached into the cementum but rarely to alveolar bone.
- Elastic-like fibres (not collagen) running longitudinally, around blood vessels.

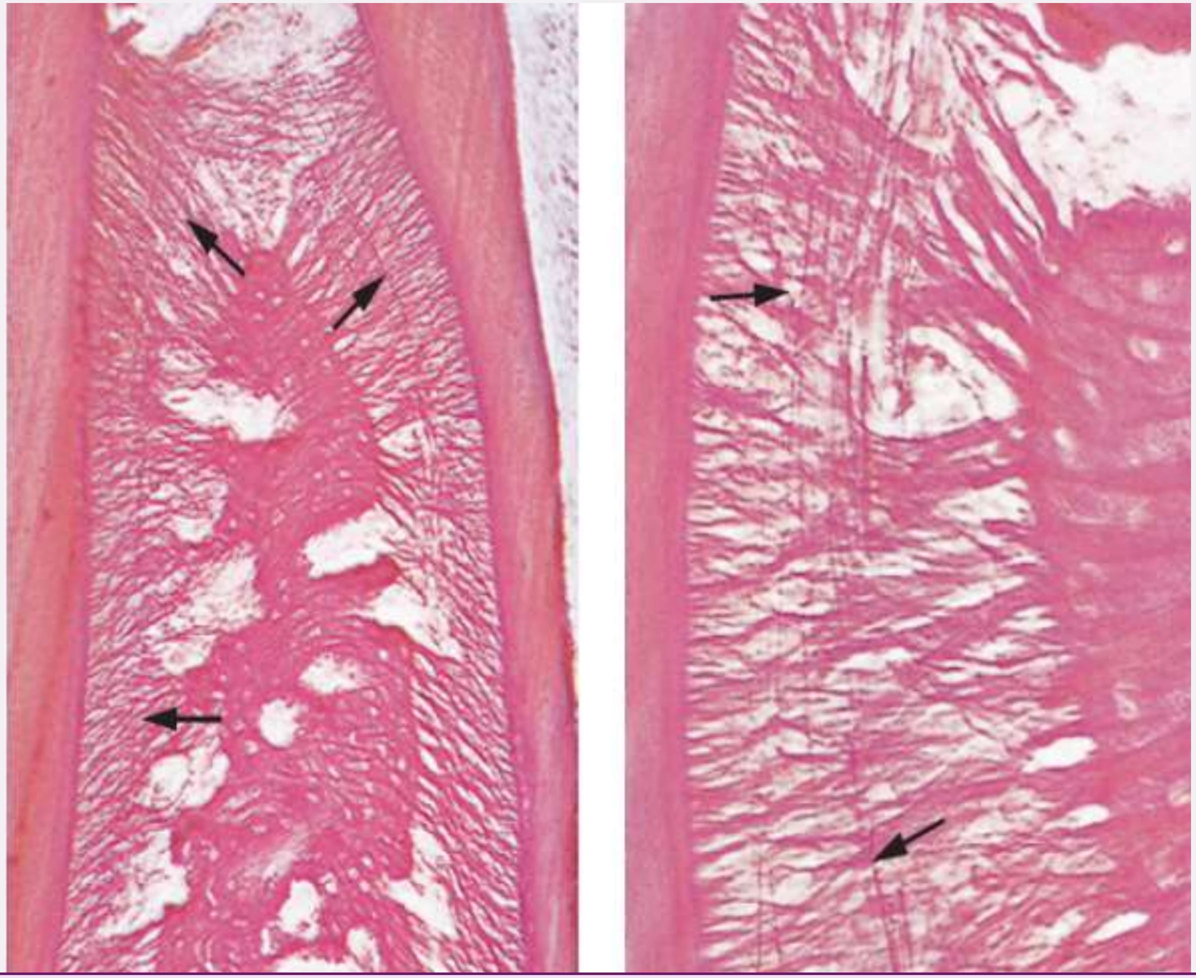


The course of oxytalan fibres (arrowed) (decalcified, longitudinal section)

OXYTALAN FIBERS

- **In the cervical region, within the pdl they tend to be more longitudinally oriented,**

- **They carry abnormally high masticatory loads.**



The course of oxytalan fibres (arrowed) (decalcified, longitudinal section)

Elaunin Fibers

•-It is a component of elastic fibres found in PDL.

-Elastin-positive, found primarily around the blood vessels in the apical region of PDL

-Provide mechanical protection for the vascular system.

Reticulin Fibers

-They are type III collagen.

-Within PDL, they crosslink to form a fine meshwork for tissue support, around collagen bundles (near alveolar bone)

-Structural integrity and flexibility

GROUND SUBSTANCE

GROUND SUBSTANCE

The turnover rate of the ground substance of PDL is faster than that of other collagenous tissues (the half-life of collagen is 3-23 days).

The components of the periodontal ligament ground substance are secreted by fibroblasts.

Helps maintain tissue fluid pressure for tooth support and the eruptive mechanism.

GROUND SUBSTANCE

Functions include ion and water binding and exchange, control of collagen fibrillogenesis, fibre orientation and binding of growth factors).

Tissue fluid pressure is high, about 10 mm Hg and the tissue fluid has been implicated in the tooth support and eruptive mechanisms.

Table 12.1 Ground substance of periodontal ligament

Glycosaminoglycans (GAGs)	Mainly hyaluronan
Proteoglycans (PGs)	Proteodermatan sulphate PG
	Chondroitin sulphate/dermatan sulphate hybrids (PG1)
Glycoproteins	Fibronectin
	Tenascin
	Vitronectin

Glycosaminoglycans

•-Glycosaminoglycans are located near the surfaces of the collagen fibrils.

-Control ionic and water exchange to influence tissue permeability and cell motility.

•-Hyaluron has a major influence upon the permeability of the tissue and cell motility.

Proteoglycans

-Decorin proteoglycans are known to assist collagen fibrillogenesis and they increase the strength of collagen fibrils.

-Biglycan is thought to control the hydration of the extracellular matrix of a connective tissue.

Glycoprotein

-Fibronectin is thought to promote attachment of cells to collagen fibrils and are thought to be involved in cell migration and orientation.

CELLULAR COMPONENTS

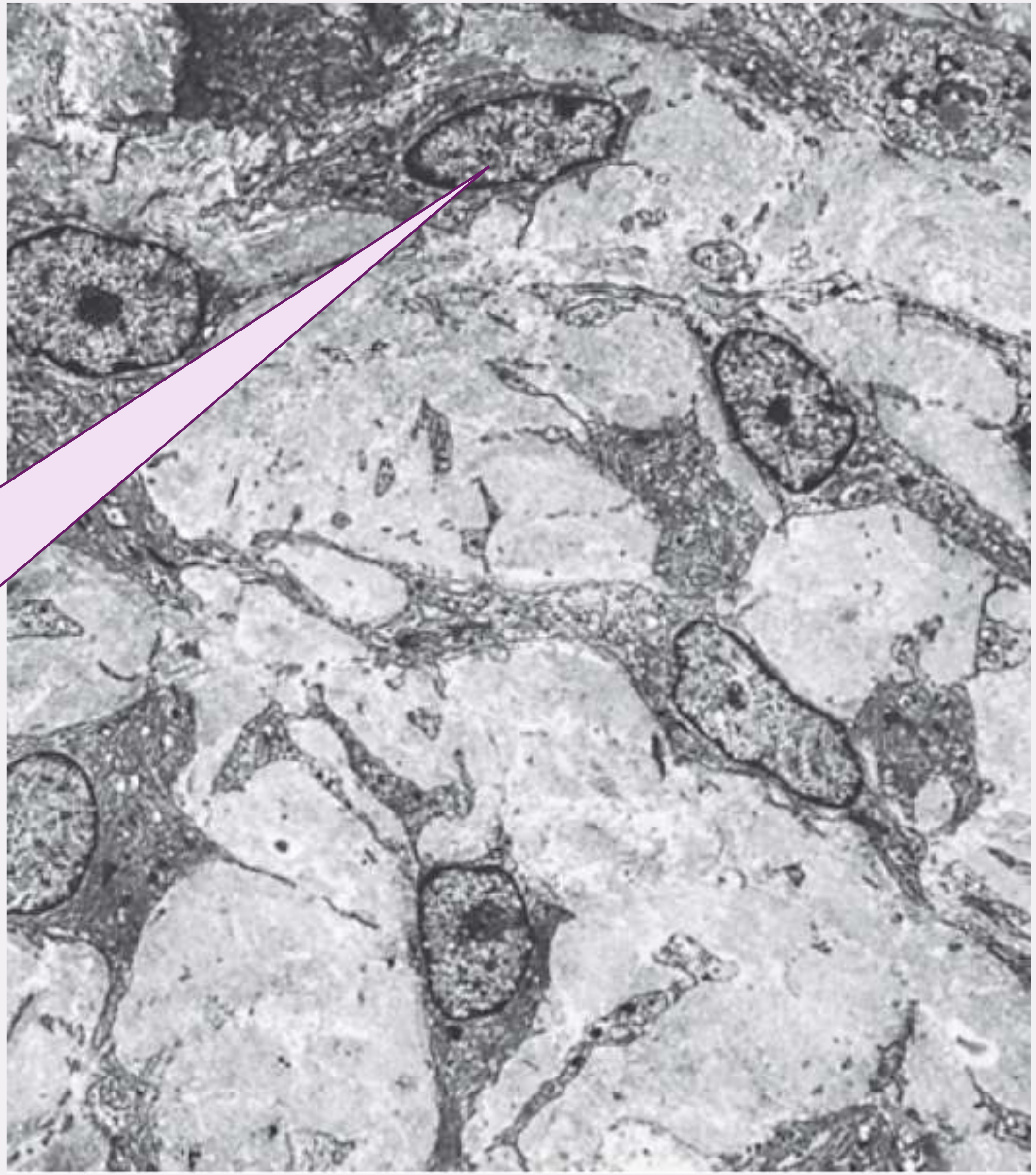
FIBROBLASTS

- **Responsible for regeneration of PDL and for tooth support in adaptive responses to mechanical loading**
- **Fusiform in shape or often appear as flattened, disc-shaped cells**
- **Active cells metabolically with high rate of protein syn**

A periodontal ligament (B) comprising numerous fibroblasts

FIBROBLASTS

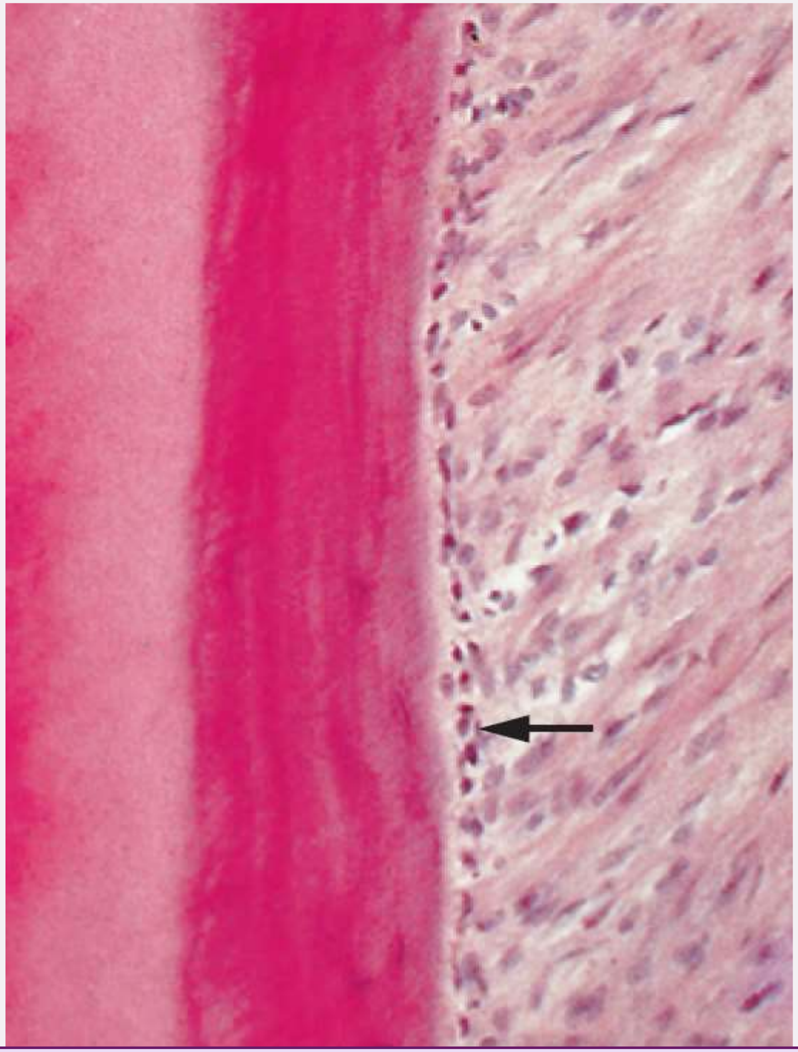
- **The fibroblasts are responsible for the synthesis and degradation of collagen**



TEM showing close association b/w principal fibres & fibroblasts of PDL

CEMENTOBLASTS

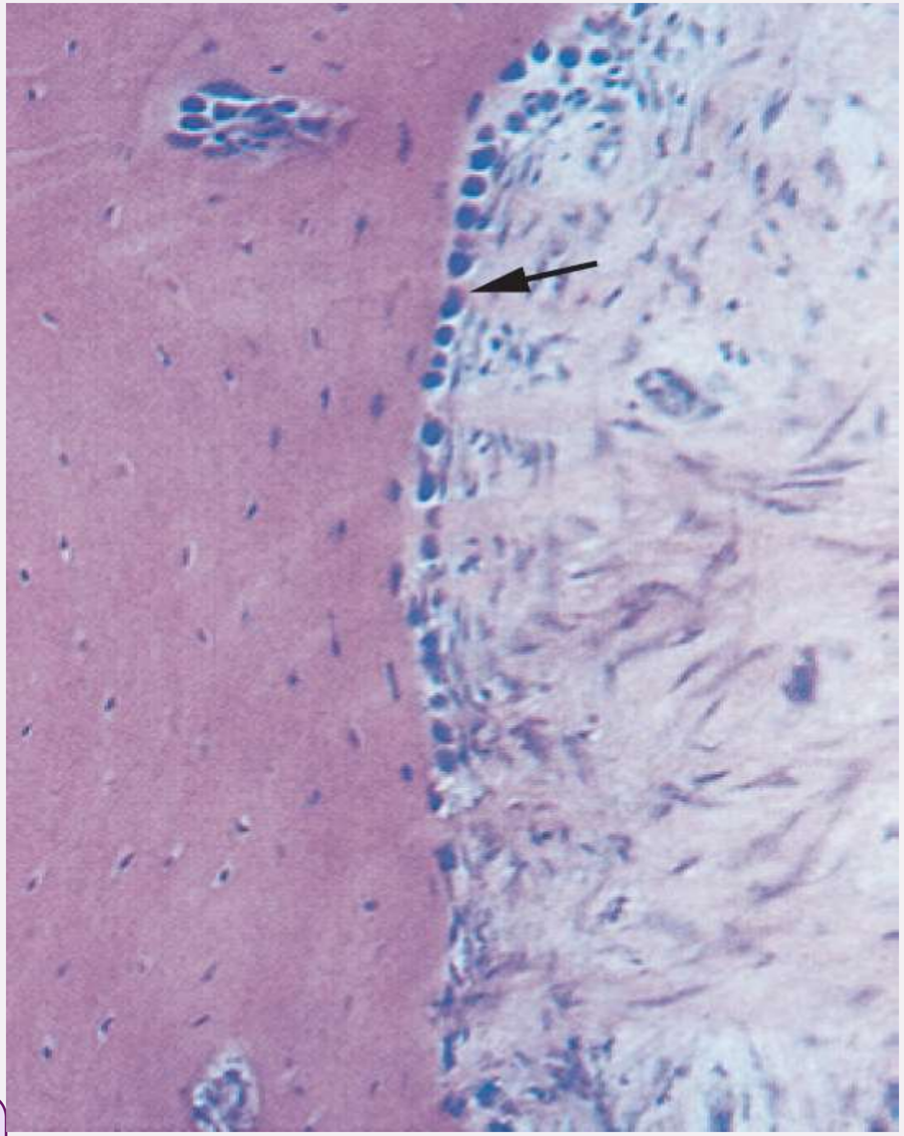
- Cementoblasts are not as elongated as periodontal fibroblasts, being squat cuboidal cells.
- They are rich in cytoplasm and have large nuclei. Like fibroblasts, they contain all the intracytoplasmic organelles necessary for protein synthesis and secretion
- The appearance of a cementoblast will depend upon its degree of activit



A layer of cementoblasts (arrowed) lining the cementum

OSTEOBLASTS

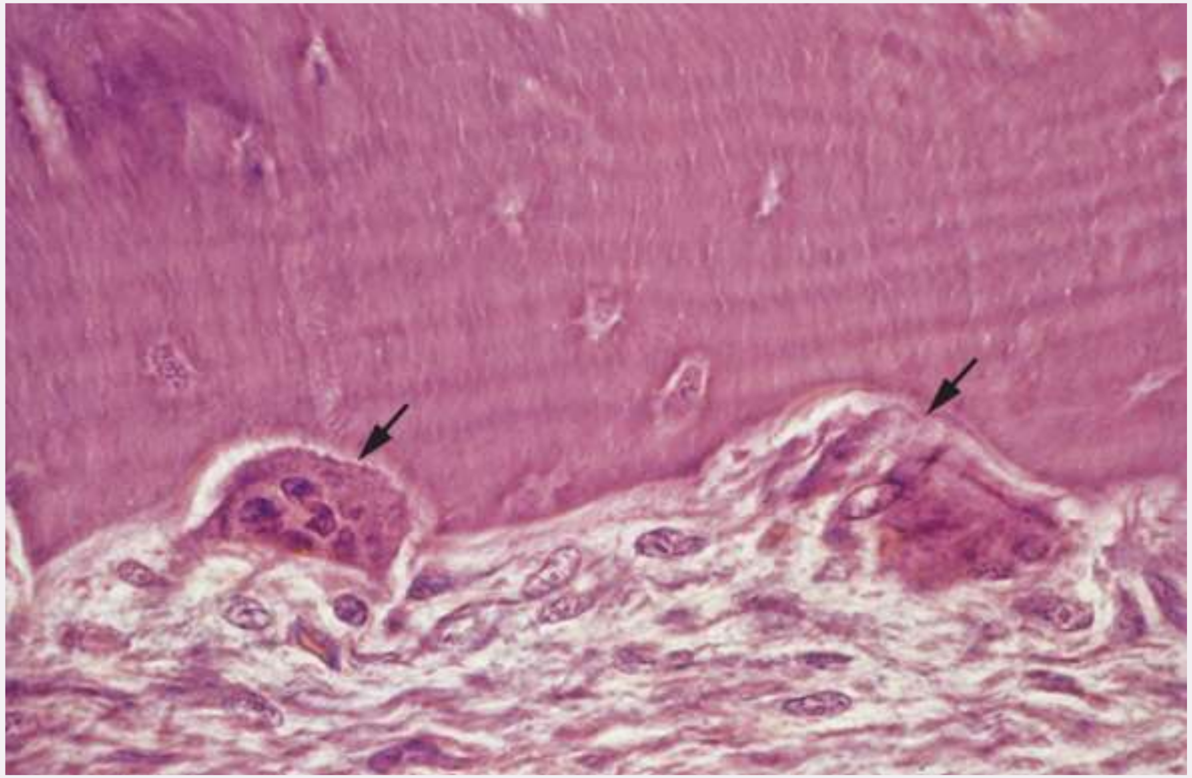
- Each osteoblast appears cuboidal and exhibits a basophilic cytoplasm that is related to the extensive endoplasmic reticulum within the cell
- They are prominent only when there is active bone formation



A layer of osteoblasts lining the alveolar bone

OSTEOCLASTS AND CEMENTOCLASTS

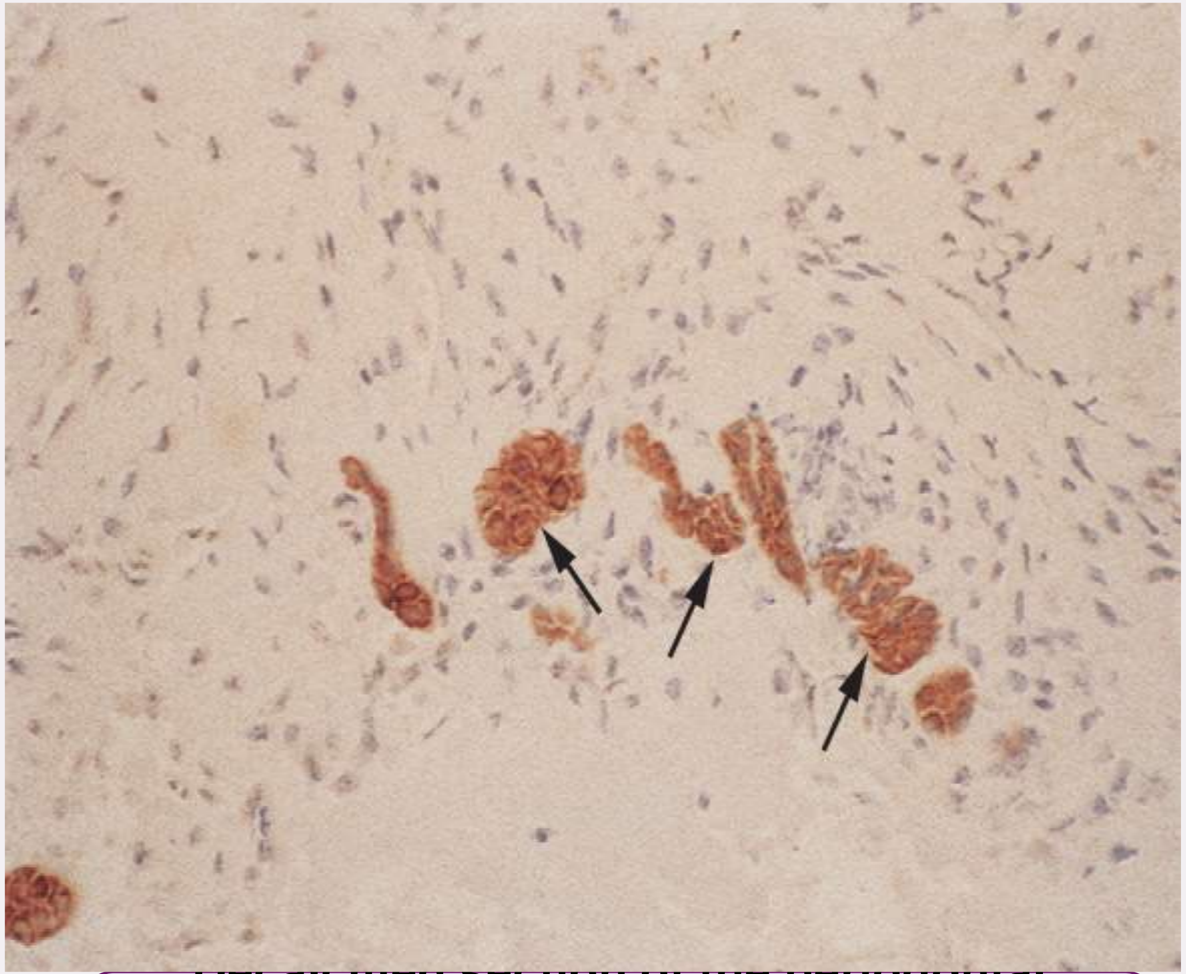
- Osteoclasts and cementoclasts (or odontoclasts) are found in areas where bone and cementum are being resorbed
- Resorption concavities termed Howship’s lacunae in which osteoclasts lie



resorbing alveolar bone and osteoclast

EPITHELIAL CELL RESTS OF MALASSEZ

- Remains of developmental Hertwig epithelial root sheath
- Clusterlike interconnecting strands parallel to the long axis of the root.
- Network of epithelial cell rests aid in maintaining the integrity of the periodontal ligament space and regeneration.



Decalcified section of the periodontal ligament showing positive brown staining for simple cytokeratins in epithelial cell rests (arrowed)

DEFENCE CELLS

Monocytes/Macrophages
- Located near the blood vessels and nerves
- Phagocytosis

Mast Cells
- Associated with blood vessels
- Responsible for production of histamine, heparin (anaphylaxis)

Eosinophils
- Sparsely present
- Phagocytosis

Dendritic Cells
- Derived from bone marrow
- Antigen presenting cells for immune response

COMPARISON WITH GINGIVAL FIBROBLASTS

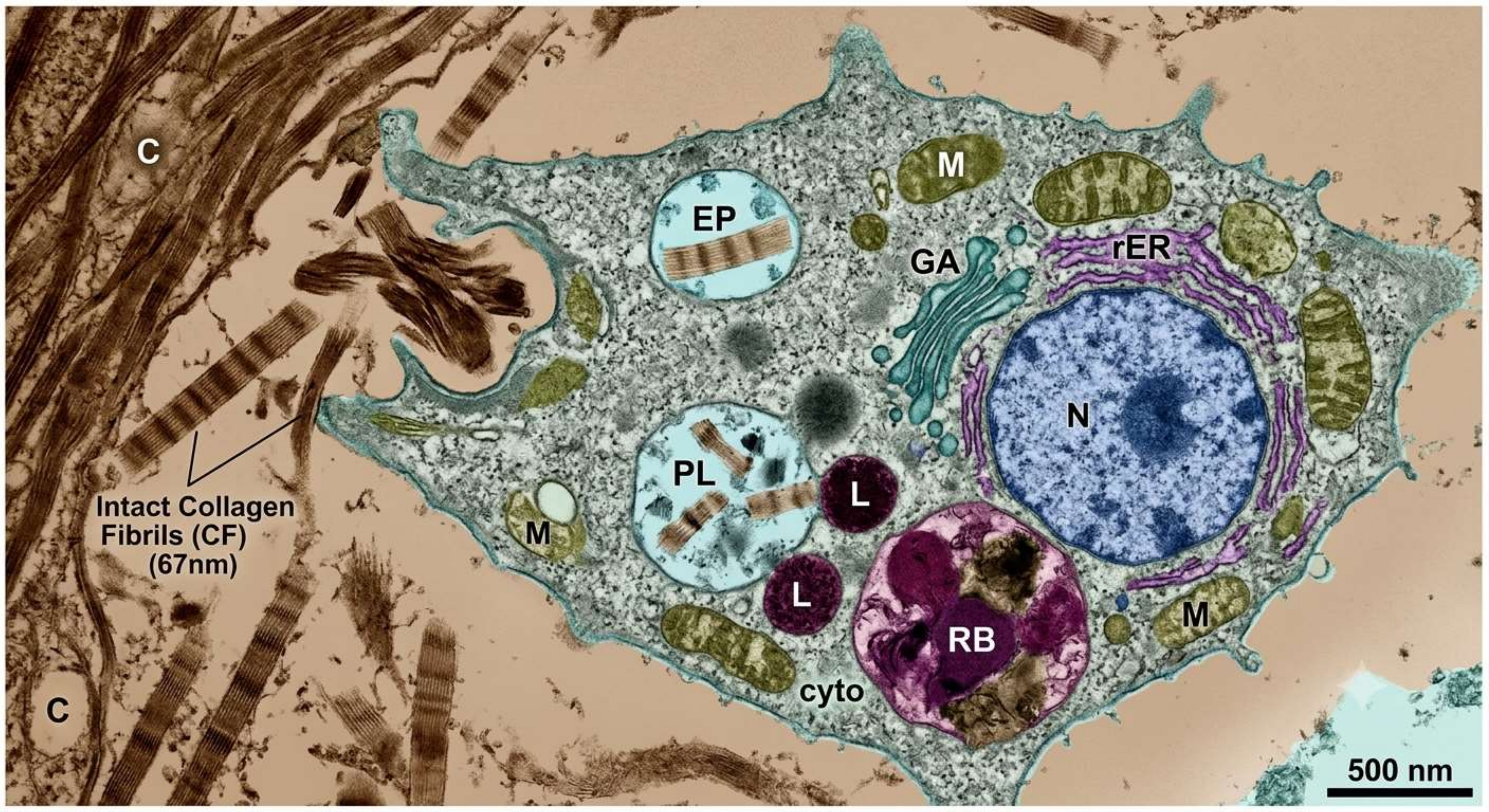
- **PDL fibroblasts are ectomesenchymal, being derived from the neural crest whereas gingival fibroblasts are mesodermal in origin.**
- **The gingival fibroblasts are less proliferative.**
- **The expression of alkaline phosphatase and cAMP is greater in periodontal ligament fibroblasts.**

DEGRADATION OF COLLAGEN

Intracellular Degradation: the fibroblasts recognize and attach to collagen fibrils. They phagocytose (engulf) portions of collagen fibrils. Under a microscope, the engulfed parts appear as ‘collagen profiles’ and are enclosed in phagosomes. Phagosomes fuse with lysosomes to form a phagolysosome and completely digest the collagen.

Extracellular Degradation: The fibroblasts in the periodontal ligament secrete MMP-1, which degrades extracellular matrix collagen under physiological conditions. To prevent uncontrolled degradation, fibroblasts also release Tissue Inhibitors of Metalloproteinases (TIMPs). Inflammation associated with periodontal disease may lead to an increased expression of MMPs and aggressive loss of collagen.

INTRACELLULAR DEGRADATION



GROWTH FACTORS AND CYTOKINES

The activities of periodontal ligament fibroblasts are modulated by numerous bioactive molecules. These molecules may be produced by the cells themselves, be produced by local inflammatory cells or be present within the extracellular matrix of the periodontal ligament or of bone and cementum.

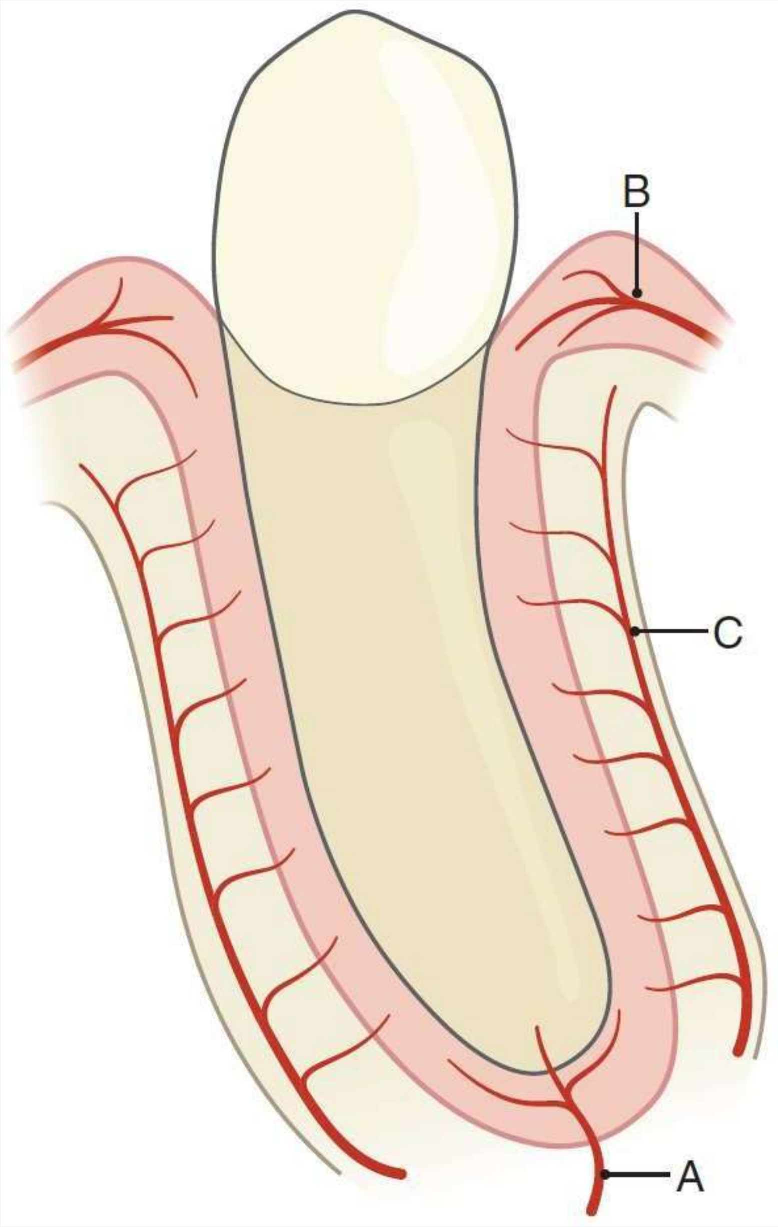
Growth Factors: Regulate cell growth, differentiation and repair.

1. PDGF: stimulates fibroblast proliferation and healing
2. TGF-β: promotes collagen and extracellular matrix production
3. IGF: supports cell growth and matrix synthesis
4. FGF: promotes fibroblast proliferation and angiogenesis

Cytokines are particularly important in inflammation and tissue remodeling. IL-1, IL-6 and TNF. They help regulate osteoclast activity and therefore bone resorption and remodeling.

ARTERIAL SUPPLY

B = arteries from gingiva (such as the lingual and palatine arteries) or interdental artery supply the coronal portion of PDL.
C = arteries from alveolar bone (vessels nearer the bone surface form a postcapillary plexus from which venules pass into the alveolar bone) or inter-radicular supplies middle and coronal part of PDL
A = arteries from dental pulp (superior and inferior alveolar arteries) supplies apical region of PDL

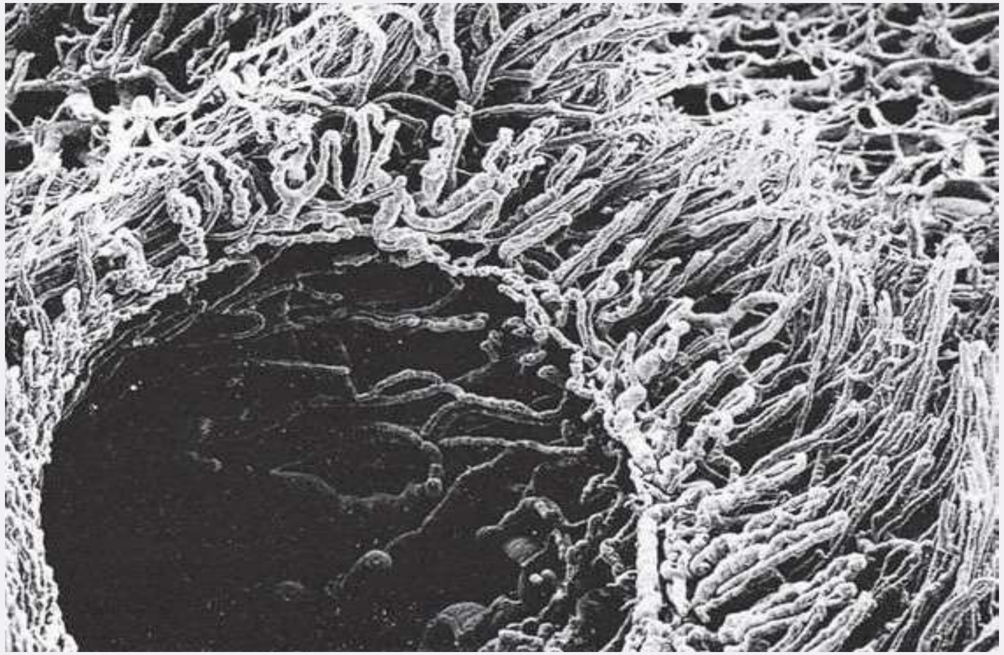


SPECIAL FEATURES RELATED TO VASCULATURE OF PDL

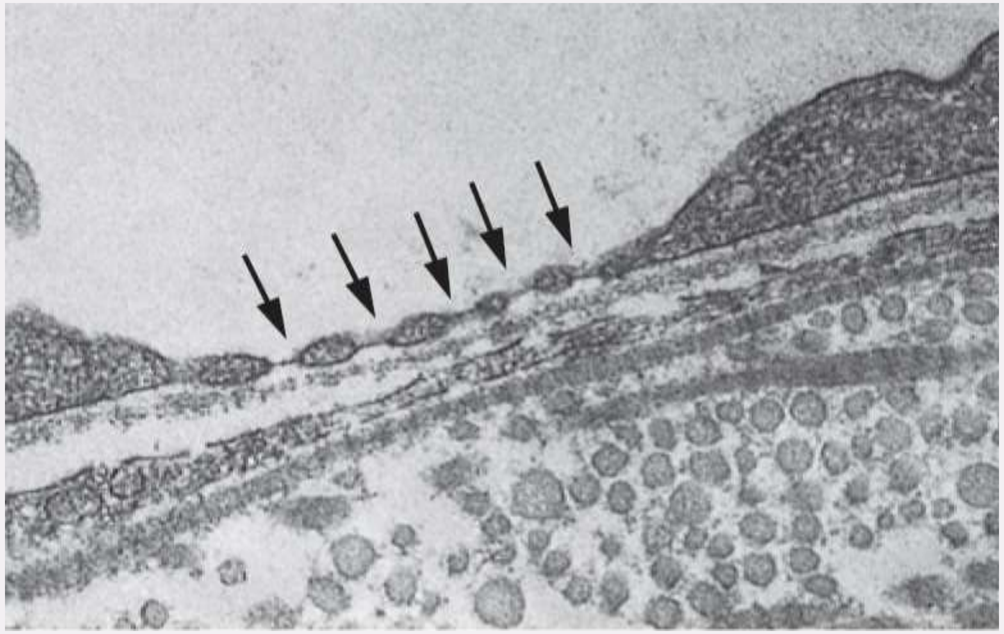
A- <u>Gingival crevicular plexus: The</u> crevicular plexus of capillary loops encircles the tooth within the connective tissue
beneath the region of the gingival crevice.

B- <u>Fenestration of capillaries:</u> The presence of fenestrated capillaries in large numbers (up to 40 × 106/mm3 of tissue).

A



B



VENOUS DRAINAGE

The veins within the periodontal ligament do not usually accompany the arteries.

1. **Intra-alveolar network: They pass directly through small perforations Volkmann’s canals in the cribriform plate and empty directly into the intra-alveolar venous networks within the surrounding spongy bone.**
2. **Gingival & apical anastomoses: a highly interconnected network (anastomoses) at both gingival and apical ends of the tooth root**

NERVOUS SUPPLY

Sensory and Autonomic Supply

The sensory fibres are associated with nociception (pain) and mechanoreception (pressure).

The autonomic fibres are associated mainly with the supply of the periodontal blood vessels.

MECHANORECEPTORS

- **These perform a major role in the transmission of touch and textural information when eating (smoothness, crunchiness, crispiness and chewiness)**
- **Periodontal ligament mechanoreceptors are majorly Ruffini-type II mechanoreceptors.**
- **Pacinian corpuscles are large, encapsulated mechanoreceptors found in the PDL. They are particularly sensitive to rapid changes in pressure or vibration.**

Receptor Type	Adaptation	Stimulus	Function
Pacinian corpuscle	Rapidly adapting	Vibration, sudden pressure	Detects rapid changes in pressure
Ruffini ending	Slowly adapting	Stretch, tension	Maintains pressure sensation, proprioception
Free nerve endings	Variable	Pain, temperature	Nociception, thermal detection

Box 12.3 **Features of the periodontal ligament indicating it is a specialised adult connective tissue**

- The principal collagen fibres have a characteristic orientation.
- The types (and amounts) of collagen (types I and III) and the variety of collagen cross-links are unlike those found within most other adult fibrous connective tissues.
- In some species, a pre-elastin-like fibre (oxytalan) is present within the periodontal ligament.
- The rate of turnover of the periodontal ligament is very fast.
- The periodontal ligament is remarkably cellular and rich in ground substance.
- The type of proteoglycan in the periodontal ligament may be specific to this tissue.
- The tissue hydrostatic pressure is high.
- The periodontal ligament fibroblasts have features unusual for fibroblasts in adult fibrous connective tissues (e.g., many intercellular contacts, cilia).
- The periodontal ligament has cells concerned with the formation of dental tissues.
- The periodontal ligament has a rich vascular and nerve supply.
- The capillaries within the periodontal ligament are fenestrated.